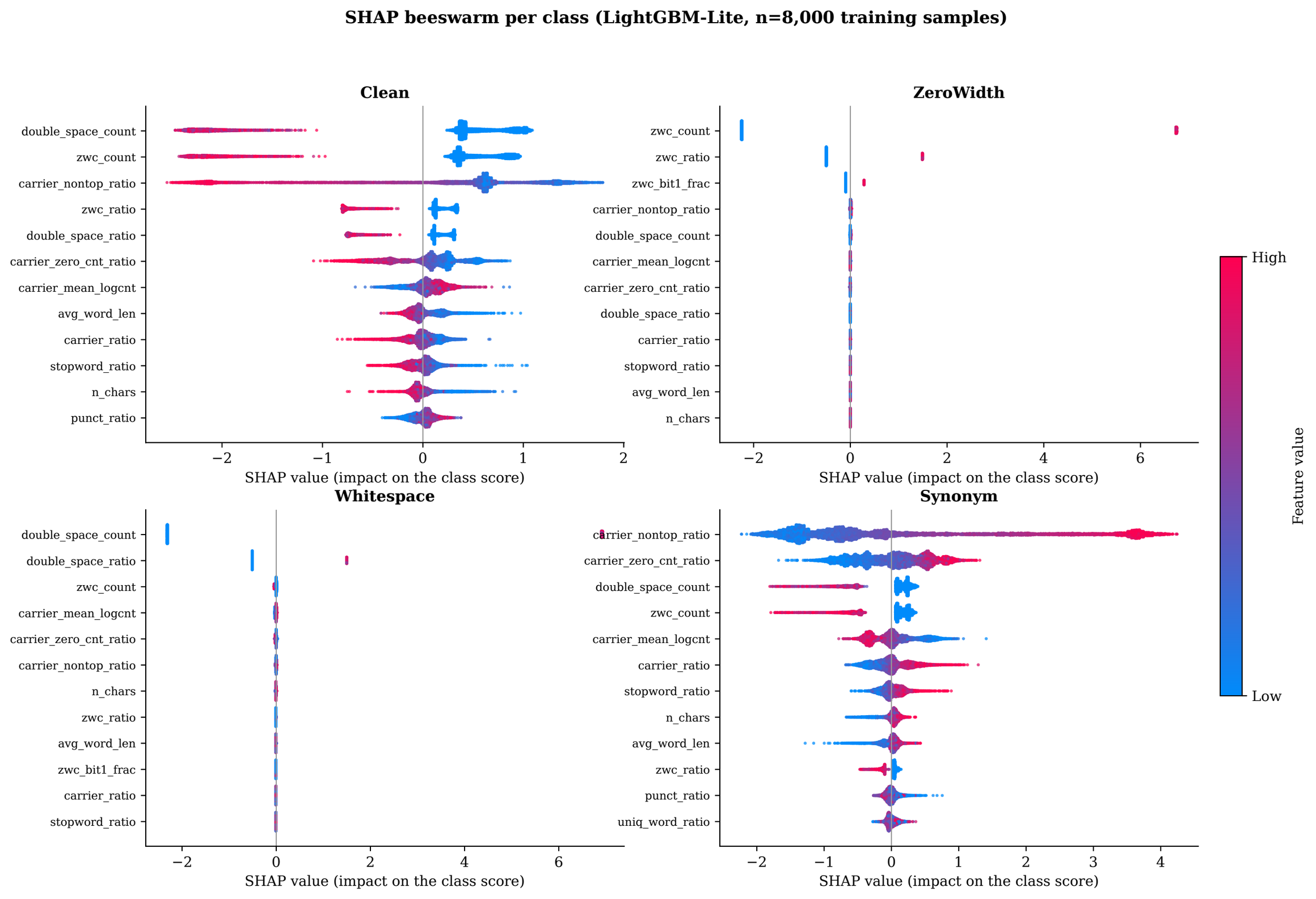

SHAP beeswarm per class (LightGBM-Lite, n=8,000 training samples)
Clean
double_space_count
zwc_count
carrier_nontop_ratio
zwc_ratio
double_space_ratio
carrier_zero_cnt_ratio
carrier_mean_logcnt
avg_word_len
carrier_ratio
stopword_ratio
n_chars
punct_ratio
−2
−1
0
1
2
SHAP value (impact on the class score)
ZeroWidth
zwc_count
zwc_ratio
zwc_bit1_frac
carrier_nontop_ratio
double_space_count
carrier_mean_logcnt
carrier_zero_cnt_ratio
double_space_ratio
carrier_ratio
stopword_ratio
avg_word_len
n_chars
−2
0
2
4
6
SHAP value (impact on the class score)
Whitespace
double_space_count
double_space_ratio
zwc_count
carrier_mean_logcnt
carrier_zero_cnt_ratio
carrier_nontop_ratio
n_chars
zwc_ratio
avg_word_len
zwc_bit1_frac
carrier_ratio
stopword_ratio
−2
0
2
4
6
SHAP value (impact on the class score)
Synonym
carrier_nontop_ratio
carrier_zero_cnt_ratio
double_space_count
zwc_count
carrier_mean_logcnt
carrier_ratio
stopword_ratio
n_chars
avg_word_len
zwc_ratio
punct_ratio
uniq_word_ratio
−2
−1
0
1
2
3
4
SHAP value (impact on the class score)
High
Low
Feature value